

Yuanjie Liu (Leonard)

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Personal website · GitHub · ORCID · arXiv · Available immediately

Objective: Plasma Theory & Simulation Researcher | Plasma Heating & Diagnostics Systems Researcher / Engineer

Education

Tianjin University of Commerce · School of Automation & Information Engineering **2020.09 – 2024.06**

B.Eng. in Automation and Artificial Intelligence

- **Relevant coursework:** Linear Algebra, Probability & Statistics, Mathematical Logic, Systems Architecture; self-study in theoretical physics and magnetic confinement fusion fundamentals.

Research Profile

- Independent researcher working on engineering-oriented magnetic confinement fusion: built HUSHFUSION Forge in Go, an end-to-end electromagnetic coil design loop (vacuum magnetostatics of magnetic-mirror configurations, equal-budget automated search, independent Python cross-check); built fusion-sim in Python / JAX, a fusion simulation workbench (TORAX / Gym-TORAX transport simulation and the gym-μ-FRC control environment); first author of an arXiv preprint (DIAP).

- Focused on theoretical modelling and diagnostics data pipelines for coupled magnetic-mirror / stellarator configurations, with strengths in magnetic configuration modelling and optimisation, transport and control simulation, numerical-experiment gating and reproducible validation, engineering implementation (Go / Python / JAX), and preprint / technical writing.

Fusion Research Projects

HUSHFUSION Forge · End-to-End Electromagnetic Coil Design *Go, Python (independent cross-check), magnetic mirrors*

- **Problem:** Coil design for fusion devices relies on manual iteration; a single experiment yields one design and the design knowledge cannot be reused.
- **Approach:** Built a closed loop "parametric design → magnetic-field physics → objective → search → design registry → design rules"; the field uses closed-form circular-loop solutions (complete elliptic integrals via AGM) cross-validated against an independent segmented Biot–Savart solver; the objective covers field strength, mirror ratio, usable field volume, ripple, cost and constraint penalties; the human baseline is a Helmholtz central cell plus mirror throat (midplane average field 1.0 T).
- **Validation:** Equal-budget comparisons show machine-designed configurations consistently outperforming the human baseline across random seeds; automated acceptance gates and cross-language oracle checks keep the results reproducible.

fusion-sim · Fusion Digital Simulation Workbench *Python, JAX, TORAX, Gymnasium*

- **Problem:** Verify that existing AI simulation stacks run locally, and determine whether in-house physics models can be cast as trainable, verifiable control problems.
- **Approach:** Track A runs TORAX (differentiable 1-D transport) + Gym-TORAX (Gymnasium control environment) + ITER ramp-up; Track B implements gym-μ-FRC v0 (μ dynamics TD1–TD21, FRC windows FC11b/FC5b, RMF cost, RL environment); clause-by-clause alignment with Lean 4 formalisation, $\mu \to 0$ limit regression, multi-seed and exhaustive reference-solution comparisons.
- **Validation:** An automated test gate covers multi-seed training, controls and exhaustive reference solutions; defect injection confirms the gate catches implementation errors, and dependencies are pinned for reproducibility.

Theoretical Physics & Formal Verification

- **Theoretical physics:** Self-taught foundations, independent research in mathematical physics and formal methods, resulting in 6 single-author preprints (2026).

- **Lean 4 formalisation:** Machine-checkable proofs in Lean 4 / Mathlib for NODS (New Object Discovery System), Hibs-Physics and YangMillsPhysics; zero sorry, zero warnings.

AI Research Workflows & Agent Systems

AI research workflows and agent systems *Go, Rust, TypeScript, Python*

- Experience in AI research workflow orchestration, multi-agent system design and AI operating systems; implemented a working fully automated multi-agent workflow system in Go covering orchestration and automatic task execution.
- Open-source projects: DIAP decentralized agent identity protocol (zero-knowledge proofs + hybrid P2P), bolloon agent resource exchange network, Hib-OS, headless.
- Other engineering work: automated simulation-data evaluation tool (Python / AutoGen, metric system and simulation data processing), multi-device parameter synchronisation and compute-scheduling framework (Rust / Python), complex-domain high-dimensional representation validation model (PyTorch).

Research & Professional Experience

Zhejiang Binjiang Research Institute · AI & Systems 2025.08 – 2026.05

Research Assistant / Research Engineer

- Participated in research on multi-agent systems and distributed computing; responsible for protocol mechanism modelling and simulation experiments.
- Carried out LLM and agent performance evaluation; designed evaluation pipelines and comparative experiments.
- Contributed to research reports and technical documentation.

GAP · Independent Research and Technical Exploration 2024.07 – 2025.08

- Independently explored AI agents and distributed systems; responsible for product planning, system architecture and core implementation.
- Designed and implemented multi-agent collaboration prototypes and a compute-scheduling system.

Publications & Output

- **First-author arXiv preprint (2025):** Yuanjie Liu, Wenpeng Xing, Ye Zhou, Gaowei Chang, Changting Lin, Meng Han. *DIAP: A Decentralized Agent Identity Protocol with Zero-Knowledge Proofs and a Hybrid P2P Stack.* arXiv:2511.11619; affiliation: Zhejiang Binjiang Research Institute.
- **Six single-author preprints (2026):** including *Counting, Not Confinement: The Yang–Mills Conjecture in the Flowing-Space Gauge Theory* and *The Sink Is a Flattening*; full list on the Publications page.
- **Open source:** hushfusion-forge (Go), fusion-sim (Python), NODS, bolloon (4 stars / 3 forks), and more; see GitHub.

Research & Technical Skills

Fusion & Simulation Magnetic-mirror coil configuration design and optimisation; vacuum magnetostatics (elliptic integrals / Biot–Savart); TORAX 1-D transport; Gym-TORAX / gym-μ-FRC control environments; FRC windows and RMF cost; equal-budget search and design registries

Numerical & Languages Go, Python, Rust; JAX / Gymnasium; numerical experiment design, equal-budget benchmarks, gating and cross-language oracle validation, reproducible experiment reports

Math & Formalisation Linear algebra, probability and statistics, mathematical logic; theoretical physics (self-taught); Lean 4 / Mathlib machine-checked proofs (zero sorry)

AI Research Systems AI research workflows, agent orchestration, multi-agent systems, AI operating systems; agent protocols (DIAP / bolloon)

Systems & Collaboration Distributed systems, multi-device parameter synchronisation, system architecture; English literature review, research reports and preprint writing